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International Journal of Advances in Engineering Sciences and Applied Mathematics

ISSN 0975-0770
Volume 12
Combined 3-4

Int J Adv Eng Sci Appl Math (2020)
12:158-170
DOI 10.1007/s12572-020-00276-4

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Linear smoothed finite element method for quasi-incompressible hyperelastic media

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Accepted: 19 November 2020 / Published online: 3 January 2021
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Abstract This work presents a linear smoothing scheme over high-order triangular elements within the framework of the cell-based strain smoothed finite element method for two-dimensional nonlinear problems. The main idea behind the proposed linear smoothing scheme is that it unlike the classical SFEM, it does not require the subdivision of the finite element cells into smoothing sub-cell. The other features of the classical SFEM are retained, such as: it does not require an explicit form of the derivatives of the basis functions, all the computations are done in the physical space, and the results are less sensitive to mesh distortion. A series of benchmark tests are done to demonstrate the validity and the stability of the proposed scheme. The validity and accuracy are confirmed by comparing the obtained numerical results with the standard FEM using quadratic triangular element and the exact solutions.

Keywords Cell-based smoothed finite element method · Hyperelasticity · Linear smoothing · Quadratic triangular element · Quasi-incompressibility

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1 Introduction

Commonly employed simplex elements in the traditional finite element method are prone to numerical difficulties such as locking (shear, membrane and volumetric), and the results are highly sensitive to the quality of the underlying background mesh. These constrains often harm the accuracy and the stability of finite element method (FEM) results when nearly- and full-incompressible material problems undergoing large deformations are considered. To alleviate aforementioned difficulties, Liu et al. [1, 2] introduced the strain smoothing approach and coined the method as the "Smoothed Finite Element Method (SFEM)." Liu et al. [3] have introduced different strain smoothing schemes, e.g. cell-based smoothing (CS-FEM), edge-based smoothing (ES-FEM) and node-based smoothing (NS-FEM) in two dimensions and face-based smoothing (FS-FEM) in three dimensions. These smoothing schemes have been widely used in various engineering fields, such as plate [4], shell [5], nearly incompressible elasticity [6, 7], and coupled with extended finite element method [8, 9]. Moreover, the selective S-FEM [10], the hybrid S-FEM [11] and enriched-S-FEM [6] were also introduced to improve the conventional S-FEM.

A notable feature of the CS-FEM is that it provides an almost identical accuracy as the FEM with simplex elements. However, Natarajan et al. [12] found that cell-based strain smoothing approach over arbitrary polytopes resulted in less accuracy than the FEM for polygonal elements. Francis et al. [13] introduced a modified cell-based smoothed-strain finite element method for arbitrary polygonal and polyhedral cells employing the linear smoothing functions based on the work of Duan et al. [14] for meshfree methods. In their work, polygons and polyhedrons are divided into triangles and tetrahedrons

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meaning the smoothing domains are constructed to compute the stiffness matrix. The transformation of quadratic serendipity shape functions introduced by Rand et al. [15] is used to obtain shape functions for polygonal cells. The Wachspress rational basis functions for polygons/polyhedra are chosen as the barycentric shape functions where the functions are converted into serendipity shape functions that satisfy the Lagrange property. To the best of author's knowledge, the extension of the smoothing schemes to higher-order simplex elements is scarce in the literature.

In the present work, the linear smoothing scheme is employed over the quadratic triangular element. Unlike the cell-based linear smoothing scheme for arbitrary polytopes or the standard CS-FEM, the proposed approach does not need to divide the finite element cells into sub-cells. Namely, quadratic triangular cells are both the target cells and the smoothing domains; thus, no further intervention to construct the sub-cells is required. The linear polynomial basis function is selected as the linear smoothing function for the CS-FEM, whereas the conventional CS-FEM has a constant linear smoothing function. Using the smoothing function, the modified shape functions are computed and the node-wise modified strain–displacement are evaluated over each smoothing domain.

The paper is organized as follows: the cell-based strain smoothing approach for the nonlinear approximation is briefly revisited in Sect. 2. In the subsequent section, the linear smoothing function in the context of CS-FEM is introduced, which illustrated the computation of the strain–displacement matrix. Then, a series of numerical tests are studied to validate the proposed method in quasi-incompressible limits (Poisson's ratio) in Sect. 4. In the final section, the performance of the proposed method is concluded and some possible future works are drawn related to the present method.

2 Nonlinear cell-based smoothed finite element method approximation

In this section, the basic idea behind cell-based-smoothed strain finite element method (CS-FEM) is briefly recalled [3]. The main features of CS-FEM are the subdivision of a finite element cell and numerical integration. In the smoothed finite element approximation, the smoothing domain is required to compute the smooth gradients. Figure 1 illustrates the construction of the smoothing domain for the CS-FEM. Figure 1a shows how a triangular cell with vertices labelled 123 is divided into three triangular sub-cells by its centroidal point 4. These sub-cells are now considered as new domains where the smoothed strain is obtained. Then, an example of the selection of the cell-based smoothing domain, when two triangular finite element cells ($\triangle 123$ and $\triangle 234$) are neighboured, is shown in Fig. 1b. When sub-cell $\triangle 125$ is chosen as a target cell, the smoothing domain for CS-FEM is constructed as itself, i.e. $\triangle 125$. Over the smoothing domain, the strain–displacement and the stiffness matrices are constructed, viz. gradients are smoothed on each smoothing domain, and not on an element.

Another distinguishing feature is the numerical integration. In the strain-smoothing approach, the numerical integration performs on the boundaries of smoothing domains as the line integration, while it performs in the element in FEM. This procedure leads to only shape functions are required at Gauss points along the boundaries of the smoothing domains. In other words, the numerical integration is performed in the current configuration and the mapping technique, so-called Jacobian, where the basis functions are mapped from the reference to current configuration is not required. To compute the strain–displacement matrix, one Gauss point located on the middle of the smoothing domain boundary and outward normal vectors are used. Figure 2 illustrates the integration scheme in the CS-FEM with the location of the Gauss points and the outward normal vectors.

For the nonlinear CS-FEM approximation, the following smoothed infinitesimal strain tensor $\bar{\varepsilon}^{\mathrm{h}}$over smoothing domain Ω is given as [6]:

$$\forall \mathbf{x} \in \Omega_k^s, \quad \bar{\varepsilon}^{\mathrm{h}}(\mathbf{x}_k) = \int_{\Omega_k^s} \varepsilon^{\mathrm{h}}(\mathbf{x}) f(\mathbf{x}) \mathrm{d}\Omega \tag{1}$$

where a point $\mathbf{x}_k$ is located in smoothing domain and $f(\mathbf{x})$ is the smoothing function. The property of the smoothing function $f(\mathbf{x})$ is:

$$\int_{\Omega_k^s} f(\mathbf{x}) \mathrm{d}\Omega = 1, \quad f > 0 \tag{2}$$

and

$$f(\mathbf{x}) = \begin{cases} 1/A_k^s, & \mathbf{x} \in \Omega_k^s \\ 0, & \mathbf{x} \notin \Omega_k^s \end{cases} \tag{3}$$

where A_k^s is the area of the smoothing domain. Using the divergence theorem, the smoothed strain of Eq. (1) can be rewritten as:

$$\bar{\varepsilon}^{\mathrm{h}}(\mathbf{x}_k) = \frac{1}{A_k^s} \int_{\Omega_k^s} \varepsilon^{\mathrm{h}}(\mathbf{x}) \mathrm{d}\Omega = \frac{1}{A_k^s} \int_{\Gamma_k^s} \mathbf{n}(\mathbf{x}) \mathbf{u}^{\mathrm{h}}(\mathbf{x}) \mathrm{d}\Gamma \tag{4}$$

where Γ_k^s is the boundary of smoothing domain and $\mathbf{n}$ is the outward normal vector in the form of the following matrix in two dimensions:

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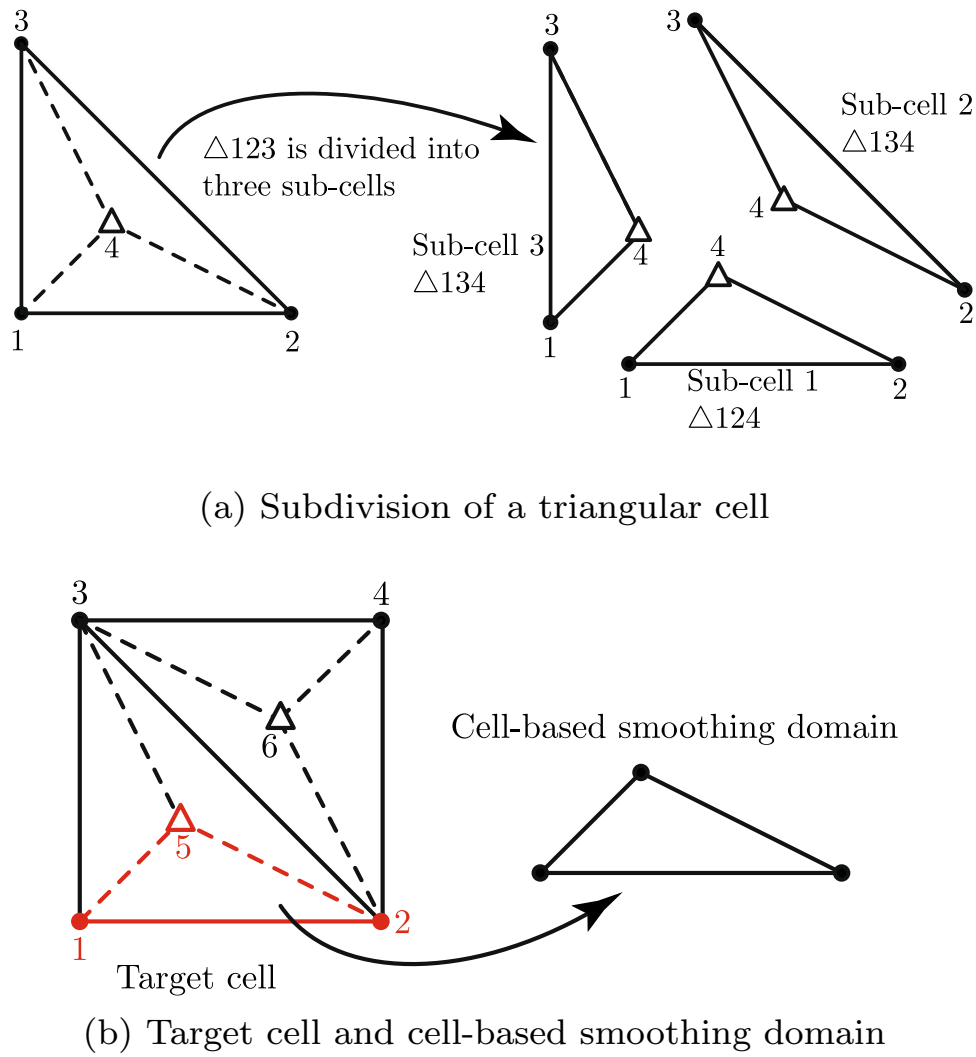

Fig. 1 Construction of cell-based smoothing domain: **a** subdivision of a triangular finite element cell and **b** target cell and its smoothing domain for CS-FEM

$$\mathbf{n}(\mathbf{x}) = \begin{bmatrix} n_1 & 0 \\ 0 & n_2 \\ n_2 & n_1 \end{bmatrix} \tag{5}$$

The details of the method are given by [16]. However, the brief information of the smoothed Galerkin weak form and its linearization is given as follows:

$$\int_\Omega \frac{\partial W}{\partial \bar{\mathbf{F}}}(\mathbf{X}, \bar{\mathbf{F}}(\mathbf{u})) : \nabla \mathbf{v} \mathrm{d}\Omega = \int_\Omega \mathbf{f} \cdot \mathbf{v} \mathrm{d}\Omega + \int_{\Gamma_N} \mathbf{g} \cdot \mathbf{v} \mathrm{d}\Gamma \tag{6}$$

where $\mathbf{v}$ is the set of admissible test function and $\bar{\mathbf{F}}$ is the smoothed deformation gradient evaluated over each smoothing domain. Eq. (6) can be expressed with the energy function $\mathcal{R}(\mathbf{u})$ and its directional derivatives $D\mathcal{R}(\mathbf{u}) \cdot \mathbf{u}$:

$$\begin{aligned} \mathcal{R}(\mathbf{u}) = & \int_\Omega \frac{\partial W}{\partial \bar{F}_{ij}}(\mathbf{X}, \bar{\mathbf{F}}(\mathbf{u})) \frac{\partial v_i}{\partial X_j} \mathrm{d}\Omega \\ & - \int_\Omega f_i v_i \mathrm{d}\Omega + \int_{\Gamma_N} g_i v_i \mathrm{d}\Gamma \end{aligned} \tag{7}$$

$$D\mathcal{R}(\mathbf{u}) \cdot \mathbf{u} = \int_\Omega \frac{\partial^2 W}{\partial \bar{F}_{ij} \partial \bar{F}_{kl}}(\mathbf{X}, \bar{\mathbf{F}}(\mathbf{u})) \frac{\partial r_k}{\partial X_l} \frac{\partial v_i}{\partial X_j} \mathrm{d}\Omega \tag{8}$$

where $i, j, k, l \in \{1, 2\}$ for two dimensions and W is the stored strain energy function.

To find an approximate solution to Eq. (6), the Newton–Raphson iterative method is employed. At iteration iter + 1, knowing the displacement $\mathbf{u}_{\text{iter}}$ from iteration iter, find $\mathbf{r}_{\text{iter}}$ that satisfies:

$$D\mathcal{R}(\mathbf{u}_{\text{iter}}) \cdot \mathbf{r}_{\text{iter}} = -\mathcal{R}(\mathbf{u}_{\text{iter}}) \tag{9}$$

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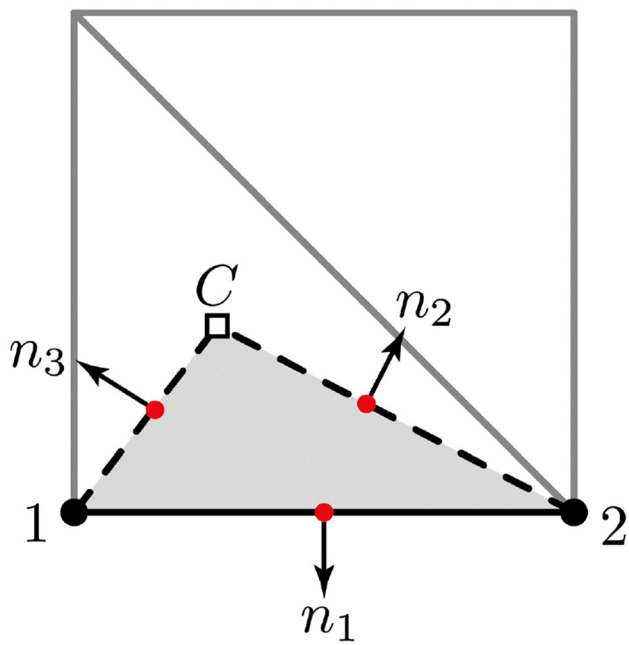

Fig. 2 The numerical integration of the strain smoothing method. Sub-cell △12C is smoothing domain of CS-FEM. White square C is a centroid of the element, and black circles are element nodes. Red circles are Gauss points located on the mid-point of the boundary of smoothing domain. Black arrows are the outward normal vectors at Gauss points

Therefore, Eqs. (7) and (8) can be rewritten as follows:

$$\mathcal{R}(\mathbf{u}) = \int_{\Omega} 2\frac{\partial W}{\partial \bar{C}_{ij}}\bar{\mathbf{F}}_{ki}\frac{\partial v_k}{\partial X_j}\mathrm{d}\Omega - \int_{\Omega} f_i v_i \mathrm{d}\Omega + \int_{\Gamma_N} g_i v_i \mathrm{d}\Gamma \tag{10}$$

and

$$D\mathcal{R}(\mathbf{u})\cdot\mathbf{r} = \int_{\Omega}\left(\frac{\partial^2 W}{\partial \bar{C}_{ij}\partial \bar{C}_{kl}}\bar{F}_{pi}\frac{\partial v_p}{\partial X_j}\bar{F}_{sk}\frac{\partial v_s}{\partial X_l} + 2\frac{\partial W}{\partial \bar{C}_{ij}}\frac{\partial r_k}{\partial X_i}\frac{\partial v_k}{\partial X_j}\right)\mathrm{d}\Omega \tag{11}$$

where $\bar{\mathbf{C}}$ is the smoothed right Cauchy–Green deformation $\bar{\mathbf{C}} = \bar{\mathbf{F}}^{\mathrm{T}}\bar{\mathbf{F}}$.

The global system of equations at each iteration can be written as:

$$\bar{\mathbf{K}}_{\mathrm{iter}}\mathbf{r}_{\mathrm{iter}} = \bar{\mathbf{b}}_{\mathrm{iter}} \tag{12}$$

Thus, the displacement $\mathbf{u}$ is obtained by the iteration method: $\bar{\mathbf{u}}_{\mathrm{iter}+1} = \bar{\mathbf{u}}_{\mathrm{iter}} + \mathbf{r}_{\mathrm{iter}}$. The smoothed tangent stiffness matrix $\bar{\mathbf{K}} = \bar{\mathbf{K}}^{\mathrm{mat}} + \bar{\mathbf{K}}^{\mathrm{geo}}$ is given as:

$$\begin{aligned}\bar{\mathbf{K}}^{\mathrm{mat}} &= \int_{\Omega}\bar{\mathbf{B}}_0^{\mathrm{T}}\bar{\mathbb{C}}\bar{\mathbf{B}}_0\mathrm{d}\Omega = \sum_{k=1}^{N}\int_{\Omega}\bar{\mathbf{B}}_0^{\mathrm{T}}\bar{\mathbb{C}}\bar{\mathbf{B}}_0\mathrm{d}\Omega = \sum_{k=1}^{N}\bar{\mathbf{B}}_0^{\mathrm{T}}\bar{\mathbb{C}}\bar{\mathbf{B}}_0 A_k\\ \bar{\mathbf{K}}^{\mathrm{geo}} &= \int_{\Omega}\bar{\mathcal{B}}^{\mathrm{T}}\bar{\mathbf{S}}\bar{\mathcal{B}}\mathrm{d}\Omega = \sum_{k=1}^{N}\int_{\Omega}\bar{\mathcal{B}}^{\mathrm{T}}\bar{\mathbf{S}}\bar{\mathcal{B}}\mathrm{d}\Omega = \sum_{k=1}^{N}\bar{\mathcal{B}}^{\mathrm{T}}\bar{\mathbf{S}}\bar{\mathcal{B}}A_k\end{aligned} \tag{13}$$

where the smoothed 4th-order elasticity tensor $\bar{\mathbb{C}}$, N is the number of target cells, A_k is the area of smoothing domain and the second Piola–Kirchhoff stress matrix is given as:

$$\bar{\mathbf{S}} = \begin{bmatrix}\bar{S}_{11} & \bar{S}_{12} & 0 & 0\\ \bar{S}_{12} & \bar{S}_{22} & 0 & 0\\ 0 & 0 & \bar{S}_{11} & \bar{S}_{12}\\ 0 & 0 & \bar{S}_{12} & \bar{S}_{22}\end{bmatrix} \tag{14}$$

The smoothed strain–displacement matrices $\bar{\mathbf{B}}_0$ and $\bar{\mathcal{B}}$ are defined as:

$$\bar{\mathbf{B}}_0 = \begin{bmatrix}\bar{B}_{I1}\bar{F}_{11} & \bar{B}_{I1}\bar{F}_{21}\\ \bar{B}_{I2}\bar{F}_{12} & \bar{B}_{I2}\bar{F}_{22}\\ \bar{B}_{I2}\bar{F}_{11} + \bar{B}_{I1}\bar{F}_{12} & \bar{B}_{I1}\bar{F}_{22} + \bar{B}_{I2}\bar{F}_{21}\end{bmatrix}$$
$$\bar{\mathcal{B}} = \begin{bmatrix}\bar{B}_{I1} & 0\\ \bar{B}_{I2} & 0\\ 0 & \bar{B}_{I1}\\ 0 & \bar{B}_{I2}\end{bmatrix} \tag{15}$$

where the smoothed deformation gradient $\bar{\mathbf{F}}$ *is evaluated as* $\bar{\mathbf{F}} = \frac{1}{A_k}\int_{\Omega}\mathbf{F}(\mathbf{x})\Phi(\mathbf{x})\mathrm{d}\Omega$.

3 Linear smoothing function in the framework of CS-FEM approximation

In the conventional strain smoothing method, the smoothing function f is a constant, that is $f(\mathbf{x}) = 1$. This smoothing function is suitable for the strain smoothing approach with linear quadrilateral or triangular elements. However, when the quadratic elements are used in S-FEM, the given smoothing function cannot be used [13]. Therefore, to use quadratic elements, the following linear polynomial basis is chosen as the linear smoothing function:

$$f(\mathbf{x}) = \{1 \quad x_1 \quad x_2\} \tag{16}$$

and its derivative is $f_{,j}(\mathbf{x}) = \{0 \quad \delta_{1j} \quad \delta_{2j}\}^{\mathrm{T}}$.

From RHS of Eq. (1), the approximation $\mathbf{u}^{\mathrm{h}}$ can be written as a scalar function $\mathbf{u}(\mathbf{x})$ at an arbitrary sampling

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point $\mathbf{x}$ in the domain Ω with nodal parameters $\mathbf{u}_I$ and the shape functions $\Psi_I(\mathbf{x})$. This would yield the basis function derivative level with the given smoothing function as follows (further details are given in Appendix A.2):

$$\int_{\Omega_k^s} \Psi_{a,j} f(\mathbf{x}) \mathrm{d}\Omega = \int_{\Gamma_k^s} \Psi_a f(\mathbf{x}) n_j \mathrm{d}\Gamma - \int_{\Omega_k^s} \Psi_a f_{,j}(\mathbf{x}) \mathrm{d}\Omega \tag{17}$$

where Ψ is the set of the shape functions. Note that in this work, Lagrange basis functions are used as shape functions. For two-dimensional problems, Eq. (17) with the linear smoothing function and its derivatives ($f(\mathbf{x}) = [1,\ x_1,\ x_2]$ and $f_{,j}(\mathbf{x}) = [0,\ \delta_{1j},\ \delta_{2j}]$) can be expanded as follows:

$$\begin{aligned} \int_{\Omega_k^s} \Psi_{a,1}(1) \mathrm{d}\Omega &= \int_{\Gamma_k^s} \Psi_a n_1 \mathrm{d}\Gamma \\ \int_{\Omega_k^s} \Psi_{a,1}(x_1) \mathrm{d}\Omega &= \int_{\Gamma_k^s} \Psi_a x_1 n_1 \mathrm{d}\Gamma - \int_{\Omega_k^s} \Psi_a \mathrm{d}\Omega \\ \int_{\Omega_k^s} \Psi_{a,1}(x_2) \mathrm{d}\Omega &= \int_{\Gamma_k^s} \Psi_a x_2 n_1 \mathrm{d}\Gamma \end{aligned} \tag{18}$$

with the smoothing function derivatives $f_{,1}(\mathbf{x}) = [0,\ 1,\ 0]$ for $\Psi_{a,1}$ and

$$\begin{aligned} \int_{\Omega_k^s} \Psi_{a,2}(1) \mathrm{d}\Omega &= \int_{\Gamma_k^s} \Psi_a n_2 \mathrm{d}\Gamma \\ \int_{\Omega_k^s} \Psi_{a,2}(x_1) \mathrm{d}\Omega &= \int_{\Gamma_k^s} \Psi_a x_1 n_2 \mathrm{d}\Gamma \\ \int_{\Omega_k^s} \Psi_{a,2}(x_2) \mathrm{d}\Omega &= \int_{\Gamma_k^s} \Psi_a x_2 n_2 \mathrm{d}\Gamma - \int_{\Omega_k^s} \Psi_a \mathrm{d}\Omega, \end{aligned} \tag{19}$$

with the smoothing domain derivatives $f_{,2}(\mathbf{x}) = [0,\ 0,\ 1]$ for $\Psi_{a,2}$.

The numerical integration in Eqs. (18) and (19) performs on the boundary of the cell-based smoothing domain and in smoothing domain. For the boundary integration scheme, the aforementioned strain smoothing shape function is used; however, for the interior integration Lagrangian basis functions are used. Figure 3 shows the numerical integration scheme for the proposed cell-based strain-smoothed method. In the proposed method, the finite element cell does not require to be divided into sub-cells as the conventional CS-FEM. Namely, the finite element cell is the target cell and smoothing domain in this case.

Another feature that can be found in Fig. 3 is the number of Gauss points. The proposed scheme has two different Gauss point locations: the one is located on the boundaries of smoothing domains and another is the interior Gauss points. As shown in Fig. 3, three interior Gauss points and two Gauss points on the boundaries are required for the numerical integration. Using the numerical integration in Eqs. (18) and (19) we can obtain the following system of equations in two dimensions:

$$\mathbf{W}_{\text{Gauss}}\, \mathbf{d}_j = \mathbf{f}_j, \quad j \in \{1, 2\} \tag{20}$$

where the matrix form of Gauss weights $\mathbf{W}_{\text{Gauss}}$ and the vector form of RHS of Eqs. (18) and (19) $\mathbf{f}_j$ are defined as, respectively:

$$\mathbf{W}_{\text{Gauss}} = \begin{bmatrix} {}^1w & {}^2w & {}^3w \\ {}^1w\,{}^1x_1 & {}^2w\,{}^2x_1 & {}^3w\,{}^3x_1 \\ {}^1w\,{}^1x_2 & {}^2w\,{}^2x_2 & {}^3w\,{}^3x_2 \end{bmatrix} \tag{21}$$

$$\begin{aligned} \mathbf{f}_1 &= \begin{bmatrix} \sum_{k=1}^{3}\sum_{g=1}^{2} \Psi_a\left({}^g_k s\right) {}_k n_1\, {}^g_k v \\ \sum_{k=1}^{3}\sum_{g=1}^{3} \Psi_a\left({}^g_k s\right) {}^g_k s_1\, {}_k n_1\, {}^g_k v - \sum_{m=1}^{3} \Psi_a({}^m r)\, {}^m w \\ \sum_{k=1}^{3}\sum_{g=1}^{2} \Psi_a\left({}^g_k s\right) {}^g_k s_2\, {}_k n_1\, {}^g_k v \end{bmatrix} \\ \mathbf{f}_2 &= \begin{bmatrix} \sum_{k=1}^{3}\sum_{g=1}^{2} \Psi_a\left({}^g_k s\right) {}_k n_2\, {}^g_k v \\ \sum_{k=1}^{3}\sum_{g=1}^{3} \Psi_a\left({}^g_k s\right) {}^g_k s_1\, {}_k n_2\, {}^g_k v \\ \sum_{k=1}^{3}\sum_{g=1}^{2} \Psi_a\left({}^g_k s\right) {}^g_k s_2\, {}_k n_2\, {}^g_k v - \sum_{m=1}^{3} \Psi_a({}^m r)\, {}^m w \end{bmatrix} \end{aligned} \tag{22}$$

where, the coordinates of the mth interior Gauss points are defined as ${}^m r = ({}^m r_1,\ {}^m r_2)$ and their weights are given as ${}^m w$. The coordinates of gth Gauss points ${}^g_k s = \left({}^g_k s_1,\ {}^g_k s_2\right)$ and their weights ${}^g_k v$ are given at the kth boundaries of smoothing domain. The outward normal at the kth boundary of smoothing domain is given as ${}_k\mathrm{n} = ({}_k n_1,\ {}_k n_2)$. Thus, the j-th shape function derivatives vector, or solution vector, $\mathbf{d}_j$ is evaluated at three internal smoothing domain Gauss points is obtained as follows:

$$d_j = \left[\, {}^1d_j \quad {}^2d_j \quad {}^3d_j \,\right]^{\mathrm{T}} = \left[\, \Psi_{a,j}({}^1r) \quad \Psi_{a,j}({}^2r) \quad \Psi_{a,j}({}^3r) \,\right]^{\mathrm{T}} \tag{23}$$

Then, using the obtained solution vector, the modified smoothed strain–displacement matrix can be obtained as follows:

$$\bar{\mathbf{B}}\left({}^k\mathbf{r}\right) = \left[\bar{B}_1\left({}^k\mathbf{r}\right) \bar{B}_2\left({}^k\mathbf{r}\right) \cdots \bar{B}_n\left({}^k\mathbf{r}\right)\right] \tag{24}$$

where $k \in \{1, 2, 3\}$ and the nodal strain–displacement matrix evaluated at the kth interior Gauss points is expressed as:

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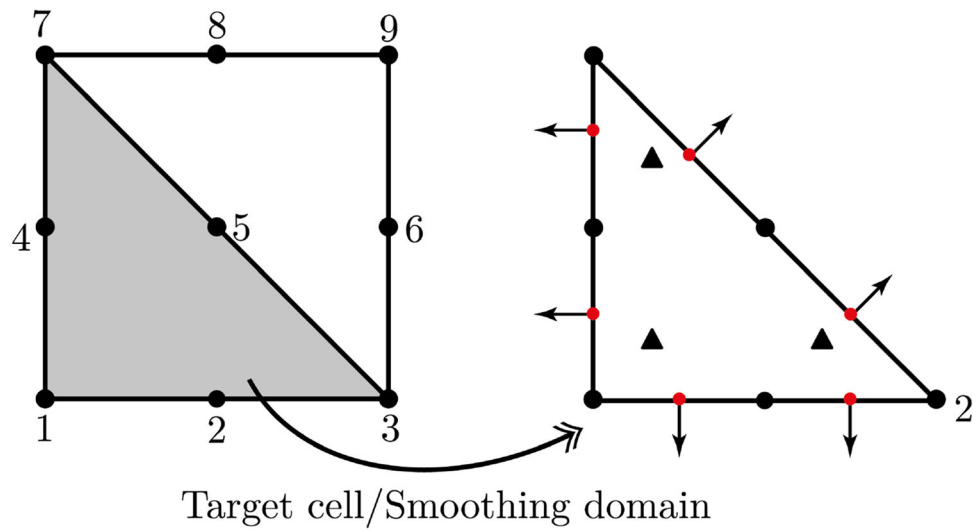

Fig. 3 The construction of smoothing domain and numerical scheme in CS-FEM with the linear smoothing function for the quadratic triangular mesh. The interior Gauss points are black triangles while Gauss points on the boundaries are red circles. The outward normal vectors are depicted as black arrows

$$\bar{B}_a\left({}^k\mathbf{r}\right) = \begin{bmatrix} {}^k d_1 & 0 \\ 0 & {}^k d_2 \\ {}^k d_2 & {}^k d_1 \end{bmatrix} \tag{25}$$

4 Numerical tests

In this section, the validity and the stability of the proposed linear smoothing function for the quadratic triangular elements in nonlinear large deformation problems are demonstrated. The following series of benchmark tests are done: (1) simple shear deformation with Dirichlet and mixed Dirichlet and Neumann boundary conditions (BCs), (2) uniform extension with lateral contraction with Dirichlet and mixed Dirichlet and Neumann BCs, (3) "Not-so-simple" shear deformation with Dirichlet BCs and (4) bending of a rectangular block with Dirichlet BCs.

For nonlinearity, the following neo-Hookean hyperelastic model is used [17]:

$$\mathrm{W} = \frac{1}{2}(\ln J)^2 - \mu \ln J + \frac{1}{2}\mu(\mathrm{tr}\mathbf{C} - 3) \tag{26}$$

where Lamé's first parameter $\lambda = \kappa - (1/2)\mu$ is defined using the shear modulus μ and bulk modulus κ in two dimensions. The Jacobian is given as the determinant of the deformation gradient $J = \det\mathbf{F}$. The present CS-FEM is compared with the exact solutions and the quadratic triangular elements in FEM. The further details of numerical tests, i.e. implementation of boundary conditions and exact solutions in strain energy, can be found in References [6, 16]. Note that all benchmark tests are considered as the dimensionless.

4.1 Simple shear deformation

First, simple shear deformation with two different types of boundary conditions is studied: (1) Dirichlet BCs and (2) mixed Dirichlet and Neumann BCs. Figure 4 depicts the initial and deformed shapes of the unit square for the problem.

To impose Dirichlet and Neumann BCs for simple shear deformation, the following deformation gradient and first Piola–Kirchhoff stress tensor are defined, respectively:

$$\mathbf{F} = \begin{bmatrix} 1 & k & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{27}$$

and

$$\mathbf{P} = \begin{bmatrix} 0 & k\mu & 0 \\ k\mu & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{28}$$

where $k > 0$ and its strain energy using Eq. (26) is given as:

$$\mathrm{W} = \frac{\mu}{2}k^2. \tag{29}$$

The current study uses $k = 1$ for the deformation gradient. To compute the strain energy, the shear modulus $\mu = 0.6$ and bulk modulus $\kappa = 100$ are used, which is equivalent to Poisson's ratio $\nu = 0.497$. Hence, the strain energy for simple shear deformation is determined to be $\mathrm{W} = 0.3$. Tables 1 and 2 provide the detailed values of displacement relative error for FEM and the proposed CS-FEM with the quadratic 6-node node triangular (T6) element. It can be

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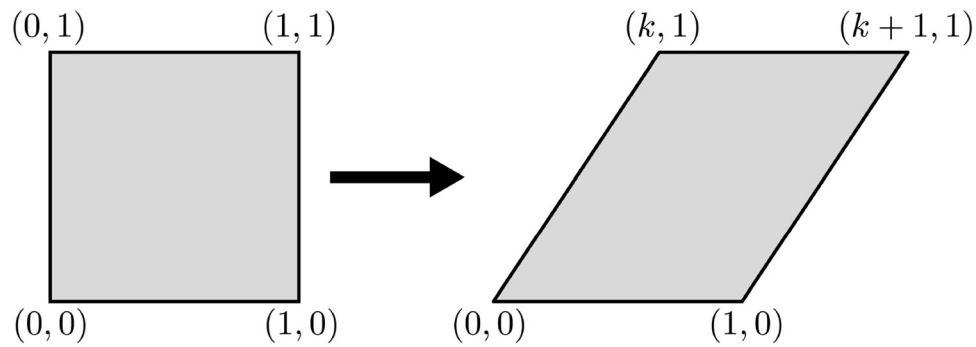

Fig. 4 Initial and deformed configurations of simple shear deformation with Dirichlet boundary conditions

Table 1 Displacement relative error ($\times 10^{-12}$) for simple shear deformation with Dirichlet BCs

DOFs	T6-FEM	T6-CS-FEM
50	0.00057	0.10846
162	0.00144	0.15607
338	0.00185	0.15717
578	0.00199	0.14690
882	0.00214	0.14206

Table 2 Displacement relative error ($\times 10^{-12}$) for simple shear deformation with Dirichlet and Neumann BCs

DOFs	FEM with T6	CS-FEM with T6
50	0.10186	0.13922
162	0.10172	0.14249
338	0.10038	0.13953
578	0.09941	0.13973
880	0.09647	0.14025

found that the proposed method shows the exact solution down to machine precision.

4.2 Uniform extension with lateral contraction

The next test is the uniform extension with lateral contraction using Dirichlet and mixed Dirichlet and Neumann BCs. The geometry and material properties used are the same as the simple shear deformation. The deformation gradient and nonzero components of the first Piola–Kirchhoff stress tensor are given as, respectively:

$$\mathbf{F} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_2 \end{bmatrix} \tag{30}$$

and

$$P_{11} = \frac{\sigma_{11}}{\lambda_1} = \mu\left(\lambda_1 - \frac{1}{\lambda_1}\right) = -P_{22} \tag{31}$$

where $\lambda_1 = 1.15$, $\lambda_2 = 1/\lambda_1$ and $\lambda_3 = 1$ are used for this test. Hence, the obtained first Piola–Kirchhoff stress are $P_{11} = -P_{22} = 0.16826087$ and the strain energy is computed as $\mathrm{W} = \frac{\mu}{2}\left(\lambda_1^2 + \frac{1}{\lambda_1^2} - 2\right) \approx 0.02359$.

Figure 5 illustrates the deformed shapes of uniform extension with lateral contraction using Dirichlet and mixed Dirichlet and Neumann BCs, respectively. It is shown in the sense that CS-FEM with the linear smoothing function provides the exactly same accuracy as the given reference solution.

Figure 6 shows the convergence of the displacement relative errors of FEM and the proposed CS-FEM with linear smoothing for the problem with Dirichlet and mixed Dirichlet and Neumann BCs. As shown in Fig. 6, a fraction is within machine precision with portion 10^{-14} for FEM and 10^{-12} for the present method.

4.3 "Not-so-simple" shear deformation

In this section and the following section, non-homogeneous deformation problems are demonstrated. Therefore, "Not-so-simple" shear deformation with Dirichlet BCs is investigated in this section. The initial and current configurations of the problem are shown in Fig. 7. As shown in Fig. 7, unlike the previous sections, a square $(0, 2) \times (0, 2)$ is used. To implement Dirichlet BCs, the following deformation gradient is defined:

$$\mathbf{F} = \begin{bmatrix} 1 & 2kX_2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{32}$$

where $k \geq 1$. The exact solution in strain energy can be calculated as $\mathrm{W} = \frac{\mu}{2}(2kX_2)^2 = 2\mu k^2 X_2^2 = 1.6$. Note that the shear and bulk moduli are the same as simple shear deformation.

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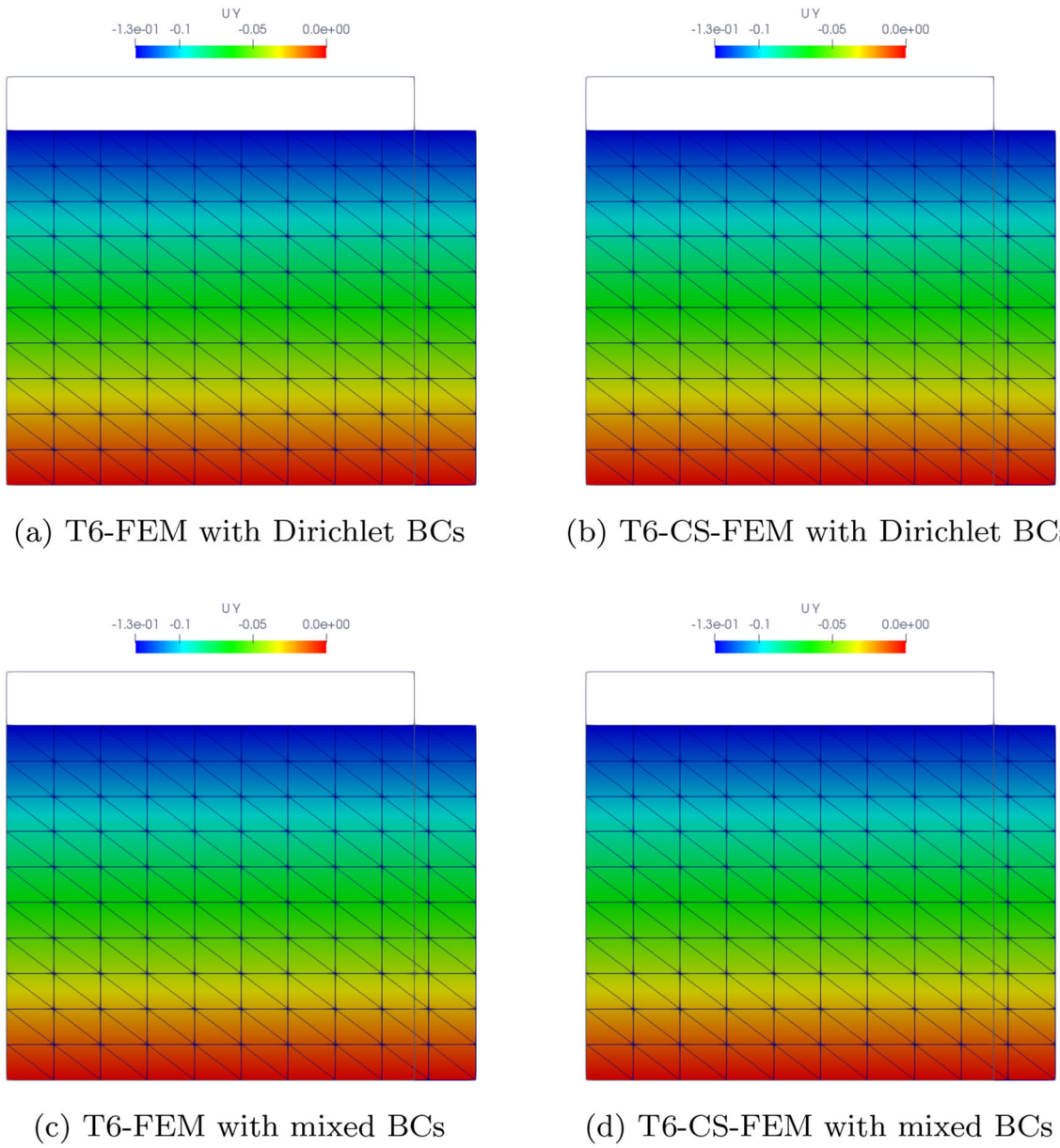

Fig. 5 Deformed configurations of uniform extension with lateral contraction with Dirichlet BCs and mixed Dirichlet & Neumann BCs: **a** FEM (T6) with Dirichlet BCs, **b** the proposed CS-FEM (T6) with Dirichlet BCs, **c** FEM (T6) with mixed Dirichlet and Neumann BCs and **d** the proposed CS-FEM (T6) with mixed Dirichlet and Neumann BCs

Figure 8 depicts the deformed shapes and horizontal displacement plots of "Not-so-simple" shear deformation with Dirichlet BCs for FEM and the proposed scheme. It can be found that the present approach provides the identical horizontal displacements as the given reference solution in displacements.

This result is also shown in Fig. 8 that the proposed CS-FEM with the linear smoothing function for the quadratic triangular element provides the accurate results when compared with traditional FEM. The relative errors in the strain energy of "Not-so-simple" shear deformation problem are illustrated in Fig. 9. It can be inferred that the results from the proposed framework are more accurate than the conventional FEM even for a coarser mesh.

4.4 Bending of a rectangular block

Lastly, a bending of a rectangular block with non-homogeneous deformation is investigated. The geometry of the block is given in Fig. 10: $A_1 = 2$, $A_2 = 3$ and $B = 2$. For this problem, shear modulus $\mu = 0.6$ and bulk modulus $\kappa = 1.95$ equivalent to Poisson's ratio $\nu = 0.36$ are used.

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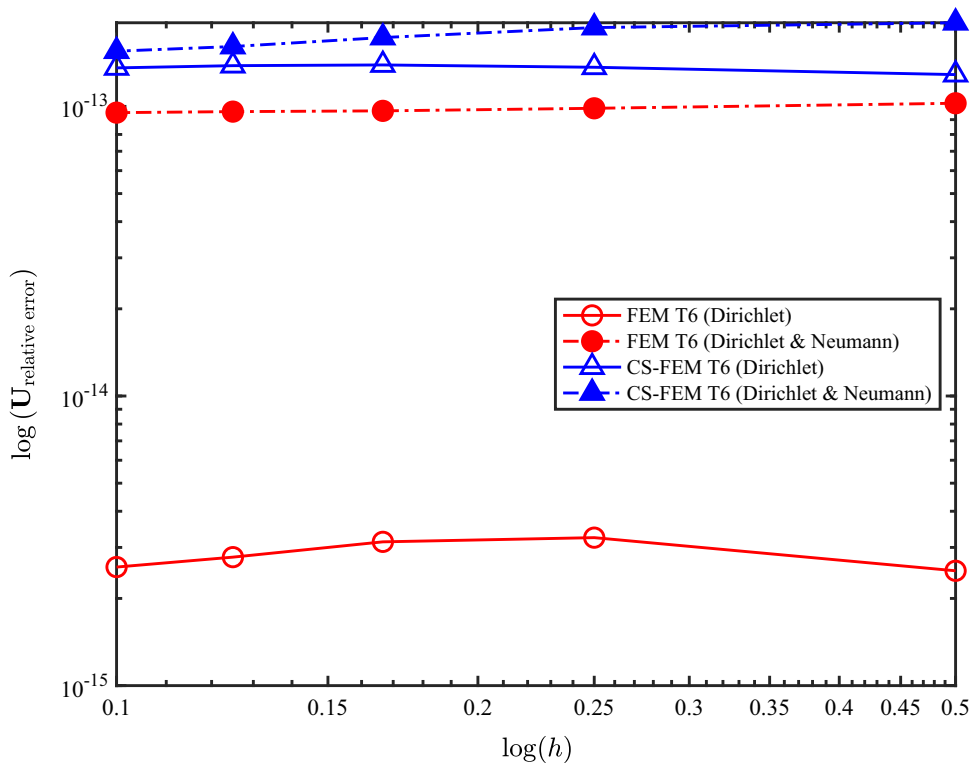

Fig. 6 Convergence of the displacement relative errors of FEM and CS-FEM for uniform extension with lateral contraction with Dirichlet and mixed Dirichlet and Neumann BCs

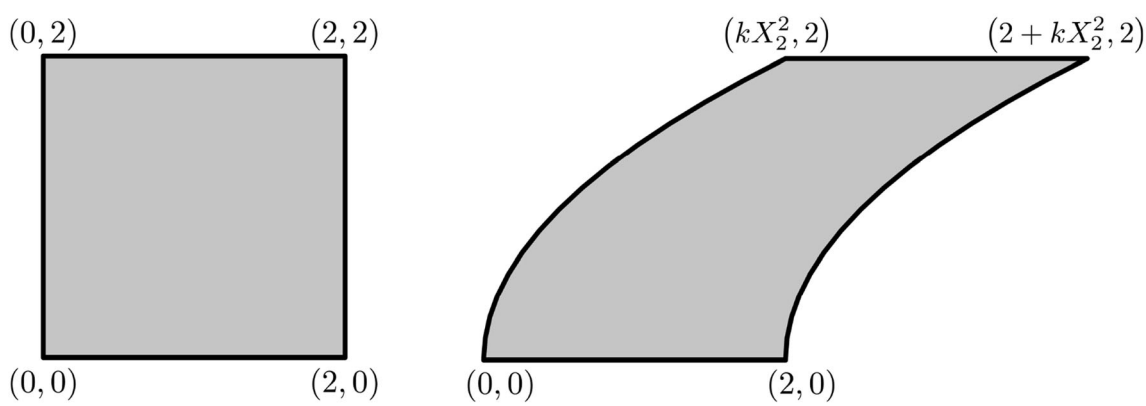

Fig. 7 The geometry and deformations of "Not-so-simple" shear deformation

To implement Dirichlet BCs, the following cylindrical coordinates in Cartesian coordinates are defined:

$$\begin{aligned} x &= r\cos\theta = \sqrt{2\alpha X}\cos\frac{Y}{\alpha} \\ y &= r\sin\theta = \sqrt{2\alpha X}\sin\frac{Y}{\alpha} \\ z &= 0 \end{aligned} \tag{33}$$

where (x, y, z) are the current Cartesian coordinates and (X, Y, Z) are the initial Cartesian coordinates. Thus, the deformation gradient for this test can be obtained as:

$$\mathbf{F} = \begin{bmatrix} f'(X) & 0 & 0 \\ 0 & f(x)g'(Y) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{34}$$

where $f(X)$, $g(Y)$, $f'(X)$ and $g'(Y)$ are:

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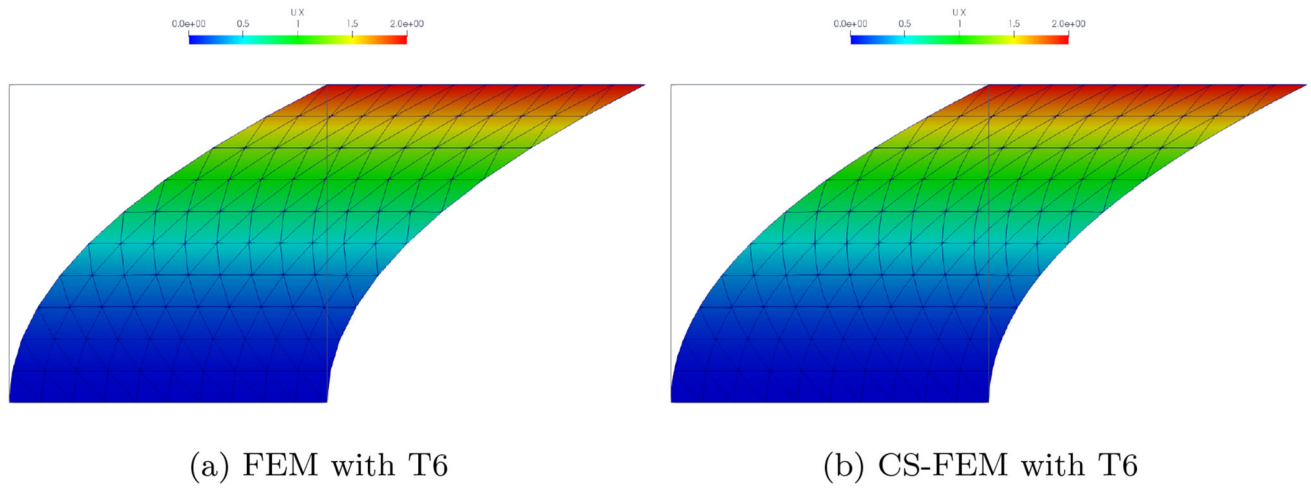

Fig. 8 Current configurations of "Not-so-simple" shear deformation with Dirichlet BCs: **a** FEM with T6 and **b** the proposed CS-FEM with T6

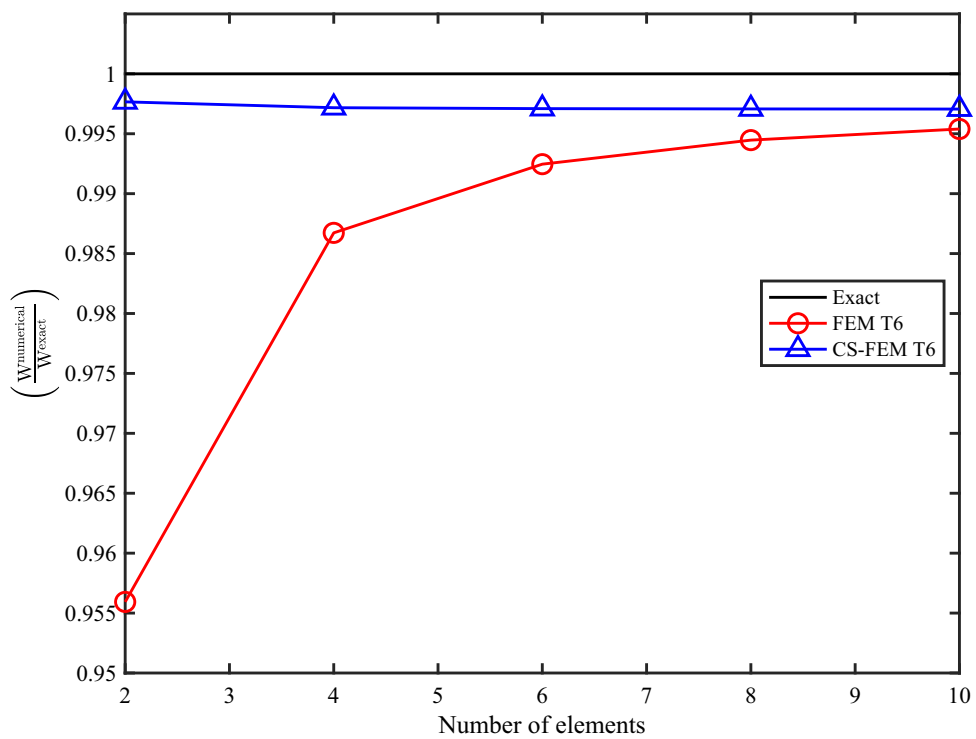

Fig. 9 The convergence of the relative error in strain energy given as $W^{\text{relative error}} = W^{\text{numerical}}/W^{\text{exact}}$

$$
\begin{aligned}
f(X) &= \sqrt{2\alpha X} \\
f'(X) &= \frac{\sqrt{2\alpha}}{2\sqrt{X}} \\
g(Y) &= \frac{1}{\alpha} Y \\
g'(Y) &= \frac{1}{\alpha}
\end{aligned}
\tag{35}
$$

where the bending factor $\alpha = 0.9$ is used in this work. Hence, the exact strain energy can be computed as follows:

$$
W = \int_2^3 \int_{-2}^2 \left\{ \mu \frac{(0.9 - 2X)^2}{3.6X} \right\} dV \approx 4.485618 \tag{36}
$$

Figure 11 illustrates the deformed shapes of the present example for FEM and the proposed linear smoothing function scheme. In this study, two cases categorized by different numbers of elements along each side are used: (1) case 1 is 2×4, 2×8, 2×12, 2×16, 2×20, 2×24, 2×28, 2×32 and (2) case 2 is 4×4, 4×8, 4×12, 4×16, 4×20, 4×24, 4×28, 4×32. The obtained results of the present method are in very good agreement

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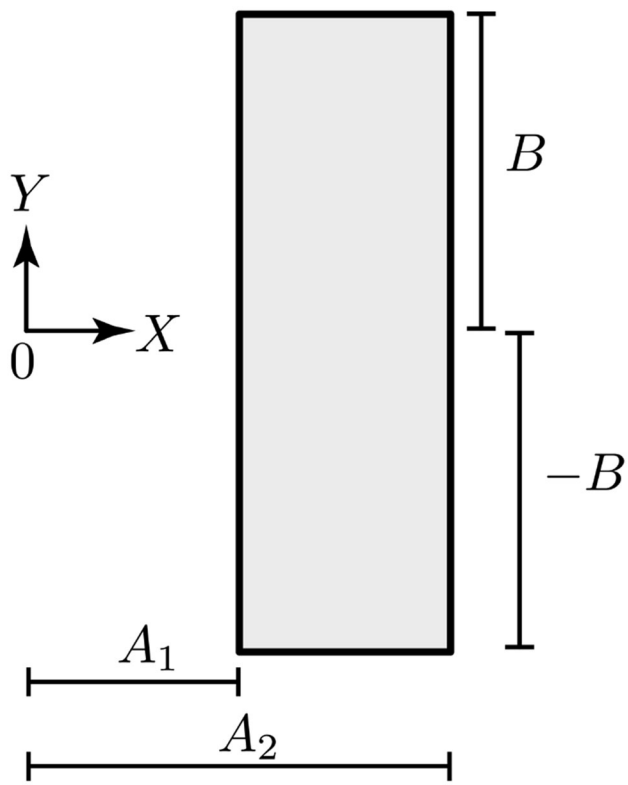

Fig. 10 The geometry of the rectangular block for bending test

with the ones in the conventional FEM and the exact solution among different numbers of finite element cells. When the bending factor α is bigger, the deformed configuration becomes a circle as shown in Fig. 11.

Figure 12 provides the convergence of the strain energy relative errors in two cases. It can be inferred that the results from the present framework are more accurate than the conventional finite element method. Moreover, the method provides more reliable and stable results for non-homogeneous deformation problems when compared to the FEM.

5 Conclusions

In this paper, the linear smoothing function is employed to the cell-based strain-smoothed finite element approximation in two-dimensional nonlinear hyperelastic media undergoing large deformations. Unlike the conventional S-FEM, the proposed scheme does not require to the division of finite element cells into sub-cells. Namely, the quadratic triangular element used the smoothing domain itself. Therefore, no further intervention for subdivision is required and this leads to an increase in the implementation efficiency for the existing S-FEM code. The present CS-FEM with the linear smoothing scheme needs two Gauss points on the boundaries of smoothing domains and three interior Gauss points in the domain. The shape functions on the boundary are evaluated by the given standard S-FEM scheme, and the shape functions inside the smoothing cells are constructed using Lagrangian basis functions. The modified smoothed shape function is then obtained by the equation of system of a set of shape functions obtained on the boundary and inside. The smoothed strain–displacement matrix and the smoothed stiffness matrix are evaluated on each smoothing domain which is in this work a finite element cell.

The robustness and the convergence properties of the CS-FEM are examined by a series of numerical tests to validate its accuracy and stability. The obtained results are compared with the exact solutions. From the numerical study, the following can be inferred:

- in the homogeneous deformation problem, the proposed CS-FEM with the linear smoothing scheme is able to yield accurate results;
- in the non-homogeneous deformation problem, the present scheme yields more accurate results than the FEM with fast convergence rate to the exact solution.

The proposed linear smoothing function will be extended to edge-based and node-based strain smoothing approximation that: (a) would effectively handle locking and (b) are less sensitive to distorted meshes than the CS-FEM, will be a topic for future communications.

Acknowledgements
by Basic Science Research Program through the National Research Foundation of Korea(NRF) funded by the Ministry of Education (2016R1A6A1A03012812).

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Appendix
The smoothed strain–displacement matrix

The following discrete trial and test functions are given using Lagrangian shape functions Ψ_I at node I:

$$\mathbf{u}^{\mathrm{h}}(\mathbf{x}) = \sum_{I=1}^{N} u_I \Psi_I(\mathbf{x}), \quad \mathbf{v}^{\mathrm{h}}(\mathbf{x}) = \sum_{I=1}^{N} u_I \Psi_I(\mathbf{x}) \tag{37}$$

Thus, from the smoothed strain of Eq. (4), the smoothed strain–displacement matrix for 2D can be evaluated as:

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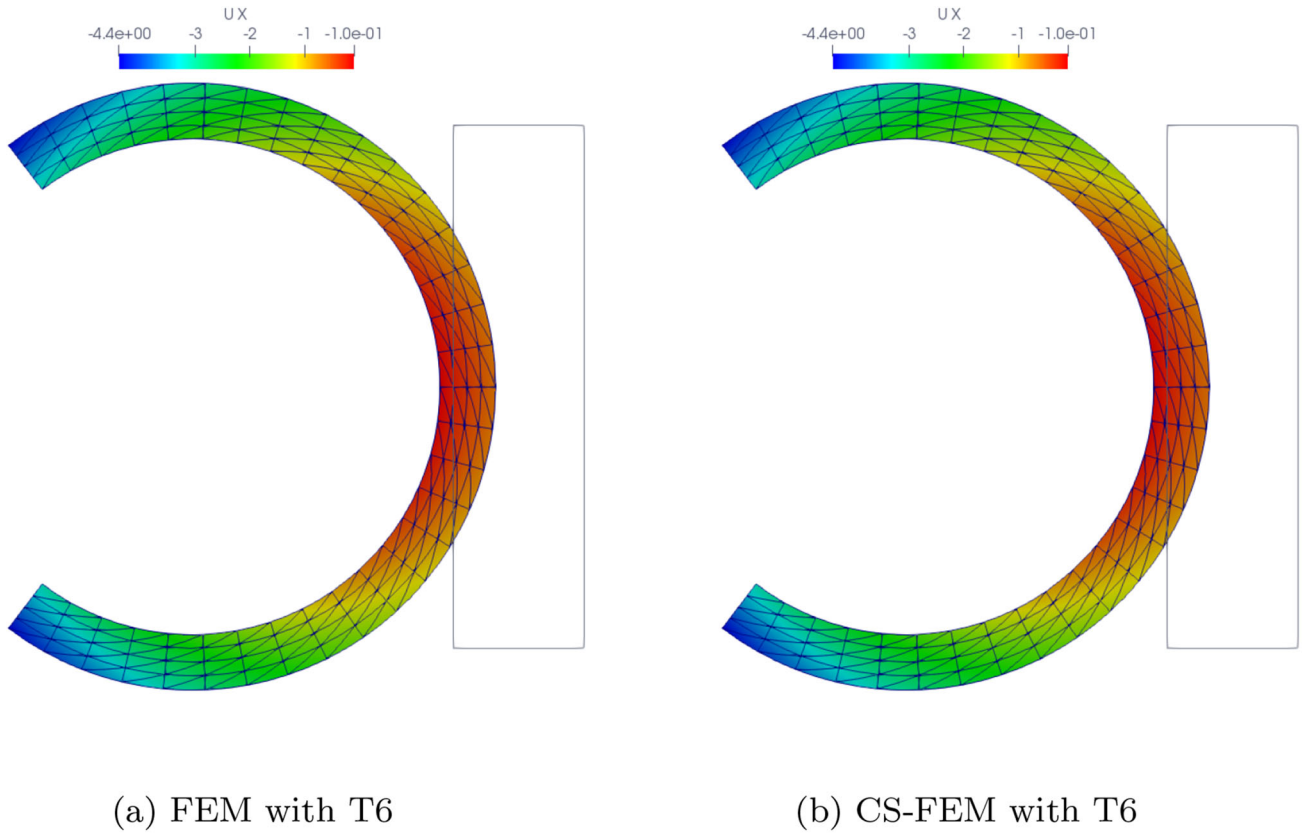

Fig. 11 Deformed shapes of the rectangular block for bending problem: **a** FEM with T6 and **b** the proposed CS-FEM with T6

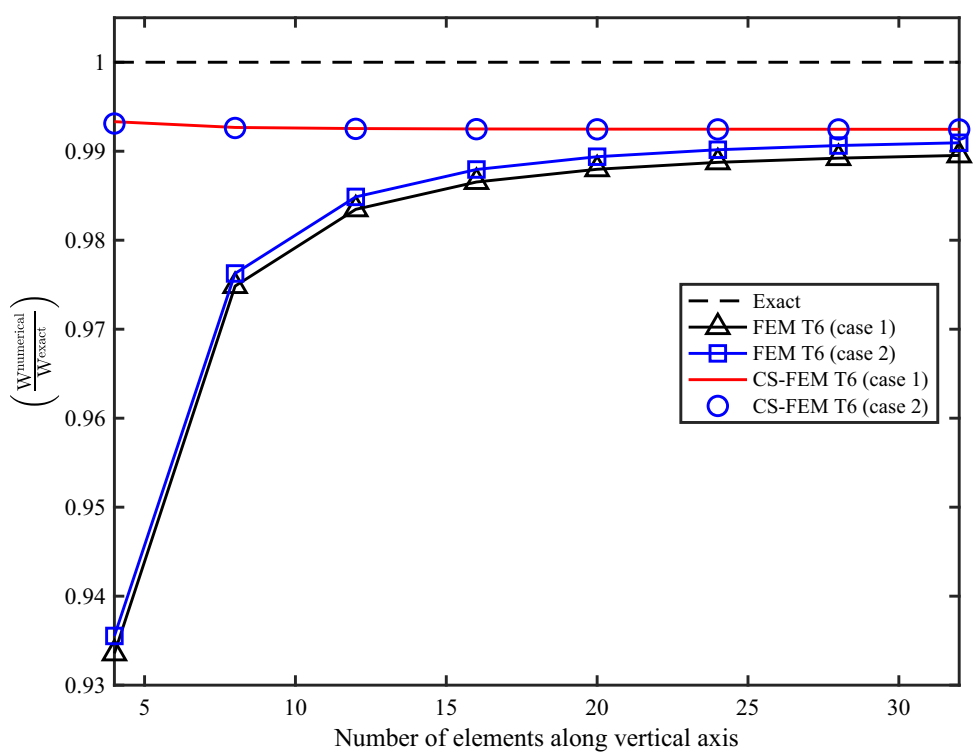

Fig. 12 The convergence of the relative error in the strain energy given as $W^{\text{relative error}} = W^{\text{numerical}}/W^{\text{exact}}$

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$$\bar{\mathbf{B}}_I(\mathbf{x}) = \frac{1}{A_k^s}\int_{\Gamma_K^s} \mathbf{n}(\mathbf{x})\Psi_I \mathrm{d}\Gamma = \begin{bmatrix} \bar{B}_{I1} & 0 \\ 0 & \bar{B}_{I2} \\ \bar{B}_{I2} & \bar{B}_{I1} \end{bmatrix} \tag{38}$$

where $\bar{B}_{Ii}$ is given as:

$$\bar{B}_{Ii} = \frac{1}{A_k^s}\int_{\Gamma_k^s} \psi_I(\mathbf{x}) n_i(\mathbf{x}) \mathrm{d}\Gamma \tag{39}$$

Consistency of the nodal derivatives

For linear smoothing approximation, as given in Eq. (16), the approximation consistency is given as [14]:

$$\sum_I \Psi_I(\mathbf{x}) = 1, \quad \sum_I \Psi_I(\mathbf{x}) X_I = x, \quad \sum_I \Psi_I(\mathbf{x}) Y_I = y \tag{40}$$

where the sampling point $\mathbf{x} = [x, y]$.

Equation (40) can be reproduced as:

$$f(\mathbf{x}) = \sum_I \Psi_I(\mathbf{x}) f(\mathbf{x}_I) \tag{41}$$

Therefore, Eq. (41) can be rewritten by taking spatial derivatives:

$$f_{,i}(\mathbf{x}) = \sum_I f(\mathbf{x}_I)\Psi_{I,a}(\mathbf{x}) \tag{42}$$

which can be extended as:

$$\sum_I \Psi_{I,1}(\mathbf{x}) = 0, \quad \sum_I \Psi_{I,1}(\mathbf{x}) X_I = 1, \quad \sum_I \Psi_{I,1}(\mathbf{x}) Y_I = 0 \tag{43}$$

for $\Psi_{I,1}$ and

$$\sum_I \Psi_{I,2}(\mathbf{x}) = 0, \quad \sum_I \Psi_{I,2}(\mathbf{x}) X_I = 0, \quad \sum_I \Psi_{I,2}(\mathbf{x}) Y_I = 1 \tag{44}$$

for $\Psi_{I,2}$. Note that, the shape functions $\Psi_I(\mathbf{x})$ need to meet a consistency condition with their derivatives $\Psi_{I,i}(\mathbf{x})$ in terms of the divergence theorem. Therefore, the divergence consistency is expressed as:

$$\int_{\Omega_S} \Psi_{I,i}(\mathbf{x}) f(\mathbf{x}) \mathrm{d}\Omega = \int_{\Gamma_S} \Psi_I(\mathbf{x}) f(\mathbf{x}) n_i \mathrm{d}\Gamma - \int_{\Omega_S} \Psi_I(\mathbf{x}) f_{,i}(\mathbf{x}) \mathrm{d}\Omega \tag{45}$$

where Ω_S and Γ_S are sub-domain and the boundaries of sub-domain under the control of the shape functions Ψ_I.

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